

MGS — Project ATLAS

Meridian Global Shipping | Excel Data Analytics Project

1. Project Overview

MGS — Project ATLAS is a practical shipping analytics project developed using **Microsoft Excel**. The project analyzes shipment operations, revenue, profitability, fuel costs, delivery performance, customers, vessels, trade lanes and CO₂ emissions.

The objective is to transform raw operational data into meaningful business insights and an executive-style dashboard.

2. Business Context

Meridian Global Shipping manages shipments across customers, vessels, ports and trade lanes.

The analysis focuses on questions such as:

- How much revenue and profit is being generated?
- Which vessels and customers perform strongly?
- Which trade lanes have higher costs?
- What are the major causes of delays?
- How is delivery performance?
- What is the company's fuel and environmental impact?

3. Dataset

- **Records:** 1,000
- **Columns:** 31
- **Tool:** Microsoft Excel

The dataset contains information related to:

- Shipment/order details
- Customers

- Vessels
- Products/cargo
- Origin and destination ports
- Trade lanes
- Revenue
- Costs
- Fuel consumption and fuel cost
- Travel duration
- Loading/unloading
- Delivery performance
- Profit
- Profit margin
- CO₂ emissions

4. Data Quality & Preparation

The dataset was checked before analysis.

Results:

- **Missing values:** 0
- **Duplicates:** 0
- **Unique shipment IDs:** 1,000

Data preparation also included checking dates, numeric values and overall consistency.

5. Excel Concepts Used

The project demonstrates practical use of:

- Sorting
- Filtering
- Excel Tables
- Conditional Formatting
- Data cleaning
- Duplicate checking
- Blank/missing-value checking
- Date formatting

- Number formatting
- Mathematical calculations
- Statistical calculations
- Logical functions
- Criteria-based functions
- Date/text functions
- Percentage calculations
- Pivot Tables
- Charts
- KPI development
- Dashboard creation
- Scenario analysis

Lookup functions such as VLOOKUP/XLOOKUP were NOT included, because they were not performed in this project.

6. Excel Formulas & Functions

Functions used included:

Aggregation

- SUM
- AVERAGE
- MIN
- MAX
- COUNT
- COUNTA

Criteria-based analysis

- COUNTIF
- COUNTIFS
- SUMIF
- SUMIFS

Logical analysis

- IF

- IFS
- AND
- OR
- IFERROR

Also:

- Arithmetic calculations
- Percentage calculations
- Date functions
- Text functions
- Rounding

7. KPI Dashboard

The project contains **8 main KPIs**:

1. **Total Revenue**
2. **Total Profit**
3. **Profit Margin %**
4. **Total Shipments**
5. **Total Fuel Cost**
6. **Average Delay Hours**
7. **On-Time Delivery %**
8. **CO₂ Emissions**

Known KPI results

Total Revenue

=SUM(P2:P1001)

Result:

58,322,468.74

Total Profit

37,740,318.60

Profit Margin

=37740318.60/58322468.74

Result:

64.71%

Total Shipments

1,000

On-Time / Completed Shipments

411

On-Time Delivery

=411/1000

Result:

41.1%

CO₂ Emissions

111,524.1

We did **not** put an invented value for Total Fuel Cost or Average Delay Hours in the documentation.

8. Business Analysis — Round 1

Questions analyzed:

Question 1

Which are the top 3 vessels by revenue?

Purpose: Identify vessels contributing the highest revenue.

Question 2

Which trade lane is the most profitable?

Purpose: Compare profitability across trade routes.

Question 3

What is the most frequent delay reason?

Purpose: Identify recurring operational problems.

Question 4

Which customer generates the highest revenue?

Purpose: Identify major revenue-generating customers.

Question 5

Which origin port is used most frequently?

Purpose: Understand shipment concentration by origin.

9. Business Analysis — Round 2

Question 1

What are the top 5 customers by profit?

Question 2

What are the top 5 vessels by profit?

Question 3

What are the top 5 trade lanes by fuel cost?

Question 4

Which customers have the highest delay hours?

The purpose was to move beyond basic totals and understand **customer, vessel and route-level performance**.

10. Scenario Analysis

The project also included practical business scenarios involving:

- Vessel maintenance
- Route/fuel cost changes
- Customer retention
- Delivery delays
- Investment decisions
- 45-day vessel unavailability

The objective was to understand how operational changes could affect business performance.

11. Pivot Table Analysis

Pivot Tables were used to summarize and compare shipment data across relevant business dimensions such as:

- Customers
- Vessels
- Trade lanes
- Delay reasons
- Financial measures
- Operational measures

They supported the business questions and dashboard analysis.

12. Charts & Visualisation

The dashboard contains **8 charts** alongside the KPIs.

The charts were used to communicate:

- Financial performance
- Vessel performance
- Customer performance
- Trade-lane performance
- Delivery performance
- Operational delays
- Cost patterns
- Environmental impact

13. Final Dashboard

The final dashboard combines:

8 KPIs + 8 charts

It provides a management-level view of:

Revenue → Profit → Shipments → Fuel → Delays → Delivery → CO₂

The purpose is to allow a business user to quickly understand operational and financial performance.

14. Analytical Workflow

The project followed this workflow:

Raw Dataset

↓

Data Quality Check

↓

Data Preparation



Excel Formulas & Functions



Pivot Table Analysis



Business Questions



KPI Development



Charts



Dashboard



Scenario Analysis



Business Insights

15. Skills Demonstrated

The project demonstrates practical knowledge of **Microsoft Excel for Data Analytics**, including:

- Data cleaning
- Data validation
- Sorting & filtering
- Conditional formatting
- Excel Tables
- Excel formulas
- Logical functions
- Criteria-based analysis
- Date/text handling
- Percentage calculations
- Pivot Tables
- Data visualization
- KPI development
- Dashboard creation

- Business analysis
- Scenario analysis

16. Project Outcome

Project ATLAS converts a raw **1,000-record shipping dataset** into a structured analytical workbook and management dashboard.

It demonstrates the complete process of taking business data from **data preparation through analysis, visualization and business interpretation**.

Portfolio information

Item	Details
Project	MGS — Project ATLAS
Company	Meridian Global Shipping
Industry	Shipping & Logistics
Tool	Microsoft Excel
Records	1,000
Columns	31
Focus	Revenue, Profit, Fuel, Delivery, Customers, Vessels, Trade Lanes, CO ₂
Dashboard	8 KPIs + 8 Charts